
STEEL DEFECT DETECTION

Cocosila-Dumitriu Rares

Manole Catalin

Grupa 334

PREZENTARE FINALTERM

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DATASET-URI FOLOSITE

- [*Severstal: Steel Defect Detection \(Kaggle\)*](#) 12.6k - Training | 5,506 - Testing

Fiecare imagine poate avea un defect, mai multe defecte, sau niciunul (defectele sunt impartite in clasa "ClassId = [1, 2, 3, 4]").

Segmentul fiecărei clase este encodat intr-un singur rand cu pixelii afectati.

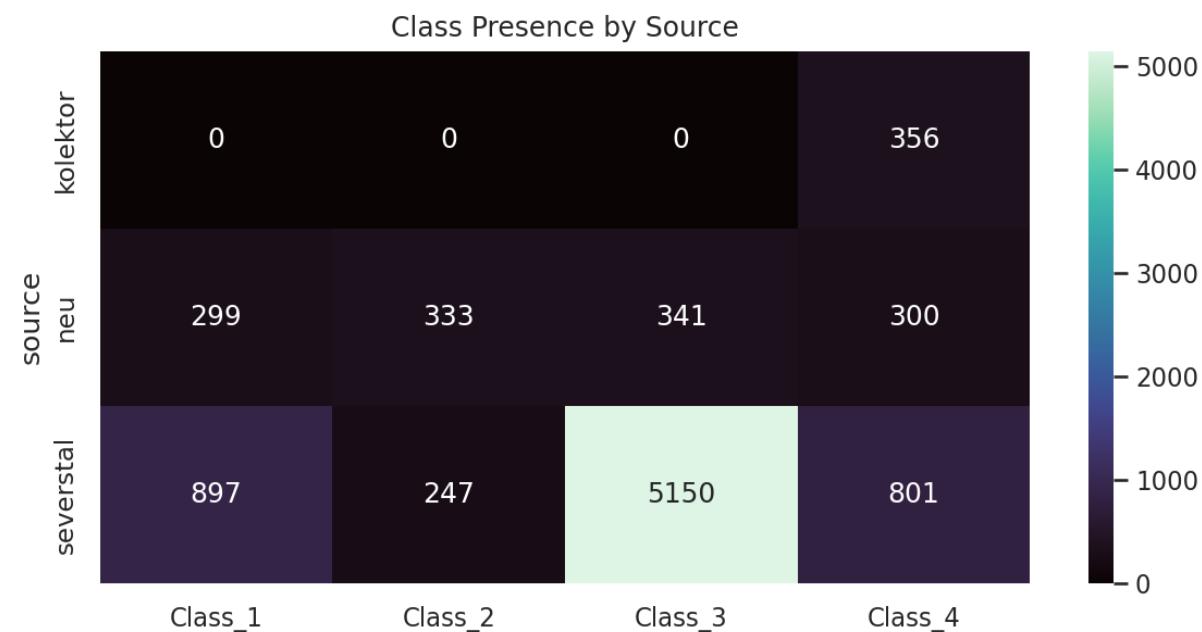
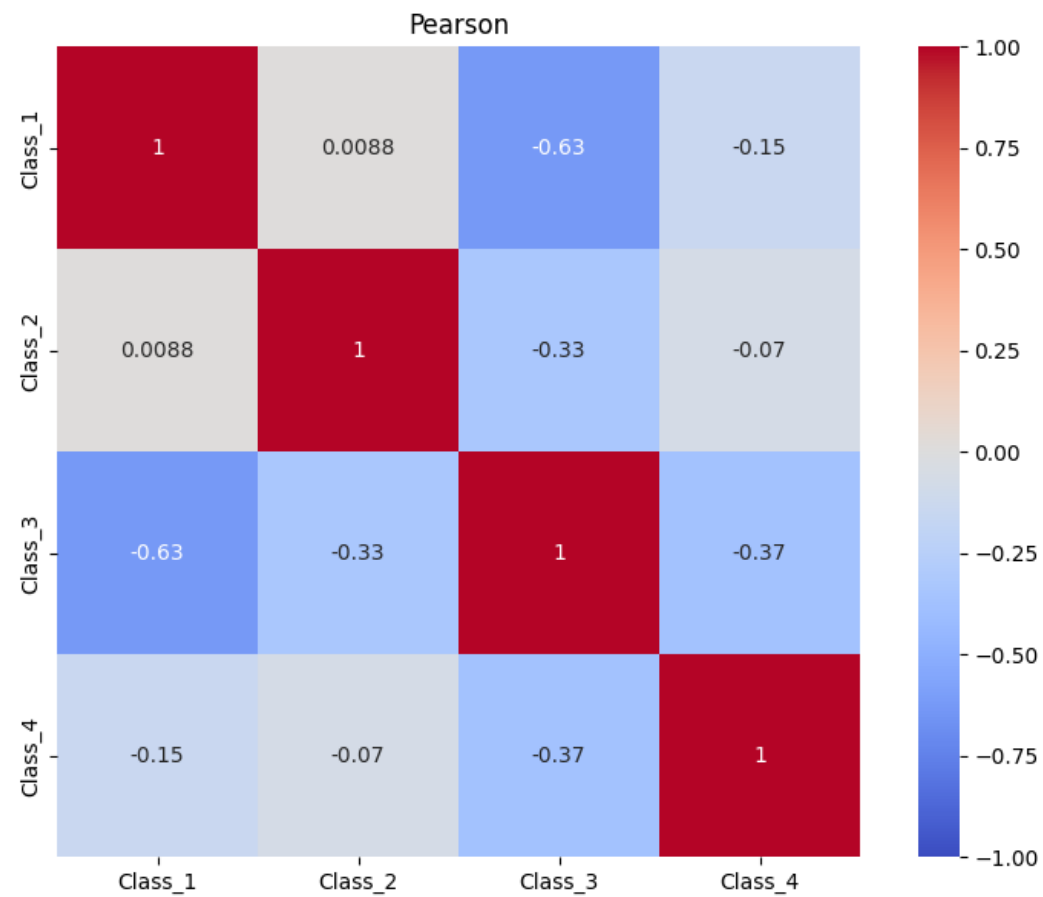
- [*NEU surface defect database*](#) 1,440 - Training | 360 - Testing

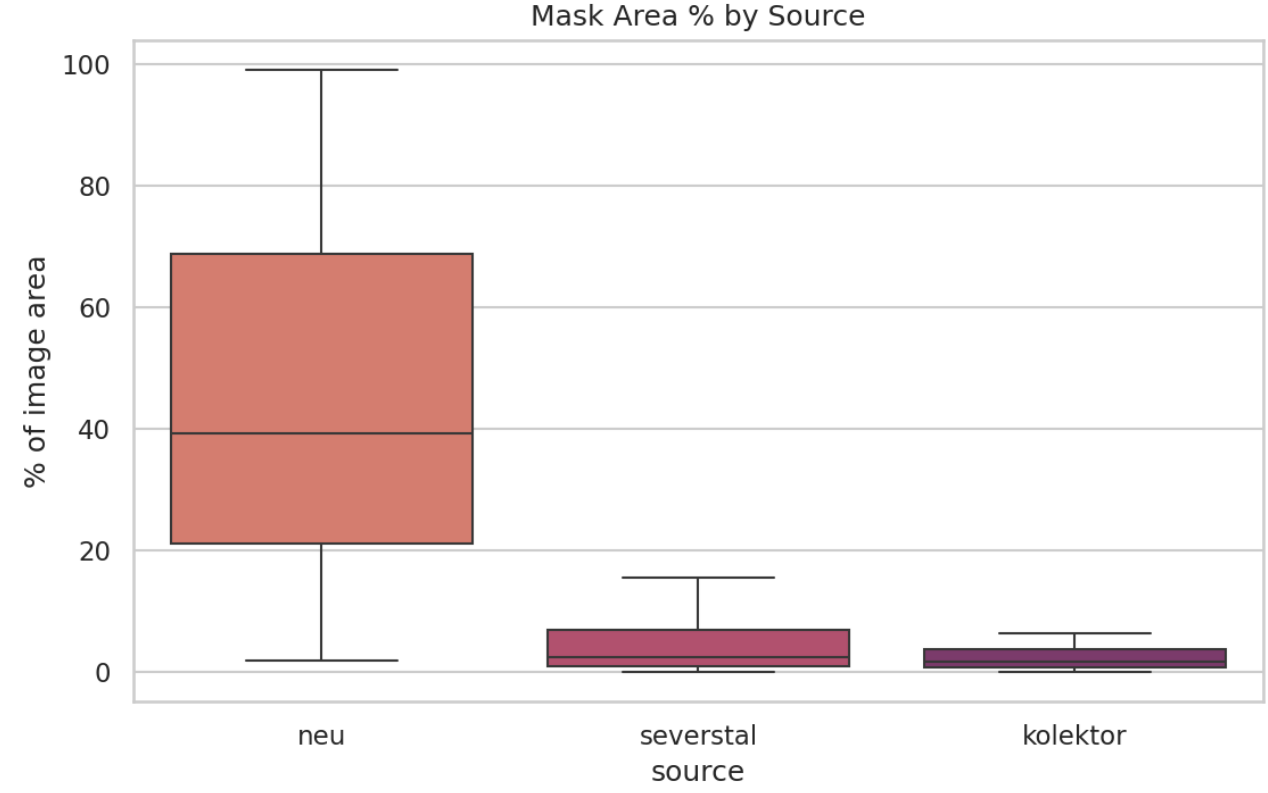
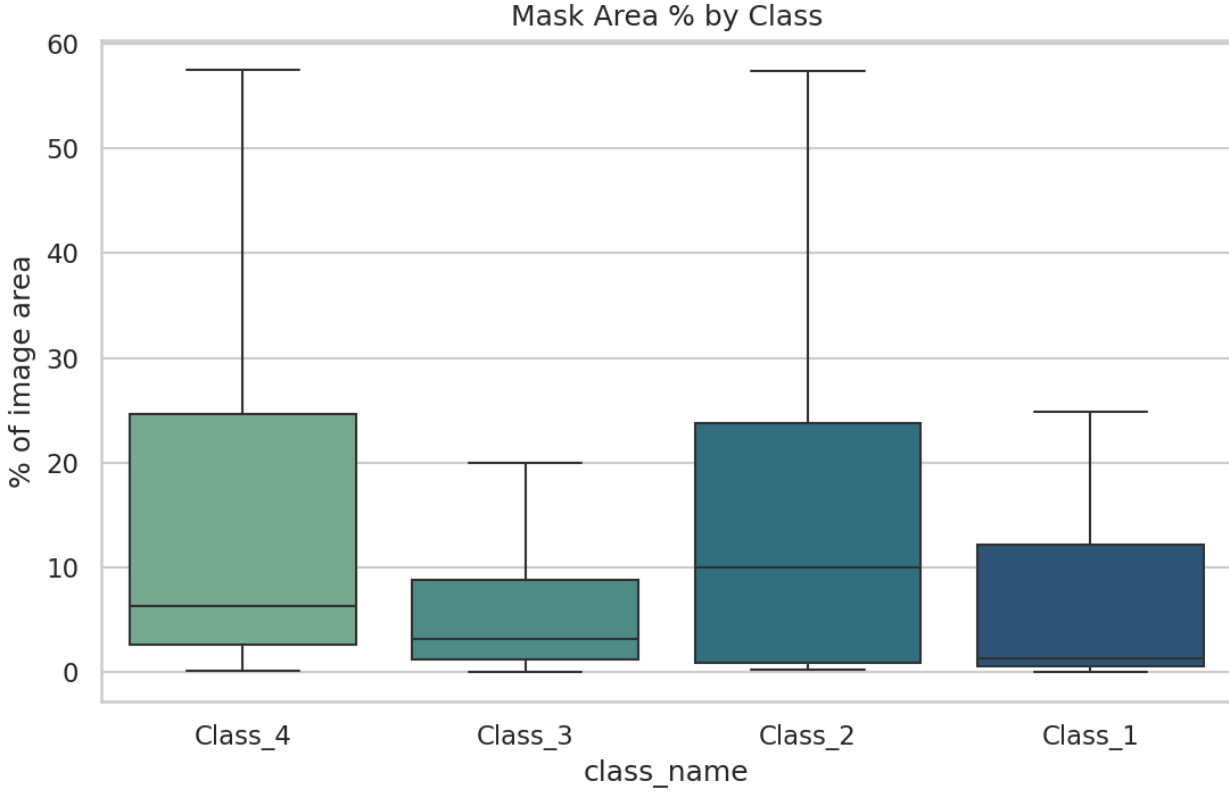
Vom avea defectele impartite in 4 clase: [1, 2, 3, 4], asemanatoare cu cele din dataset-ul de pe kaggle. Dar pentru o poza, vom avea fie un singur tip de defect, fie niciunul.

- [*KolectorSDD2*](#) 2,331 - Training | 1,004 – Testing

Dataset foarte dezechilibrat, ce contine multe imagini curate. Contine initial doar o clasa, care se aseamana cel mai mult cu clasa 4 din dataset-ul kaggle.

ANALIZA DATELOR





MODELE EVALUATE

1. U-Net - arhitectura encoder-decoder cu skip connections, folosita ca baseline eficient
2. Mask R-CNN - model de segmentare bazat pe regiuni, utilizat pentru instance segmentation
3. YOLO - model rapid, folosit pentru scenarii cu constrangeri de timp si inferenta in timp real
4. SegFormer-B0 - arhitectura bazata pe transformer care proceseaza imaginea ca secventa de patch-uri
5. SAM - bazat pe transformer encoder si mask decoder, la noi adaptat prin utilizarea encoderului inghetat si a unui head custom
6. DINOv2 - model self-supervised bazat pe Vision Transformer pentru invatarea reprezentarilor vizuale fara etichete

METRICI DE EVALUARE

- Intersection over Union (IoU) / Jaccard Index: suprapunerea dintre masca prezisa si cea reala, penalizand puternic erorile de localizare

$$IoU = \frac{\text{Area of Overlap}}{\text{Area of Union}} = \frac{TP}{TP + FP + FN}$$

- Dice Coefficient / F1 score: similaritatea intre masca prezisa si ground truth (clase dezechilibrate)

$$Dice = \frac{2 \times \text{Area of Overlap}}{\text{Total number of pixels in both}} = \frac{2TP}{2TP + FP + FN}$$

METODE DE OPTIMIZARE

- Mixed Precision (AMP) - utilizarea simultana a numerelor pe 16 si 32 de biti pentru accelerarea antrenarii si reducerea consumului de memorie
- Fast Lookups - optimizarea accesului la date prin dictionare si indexari, reducand timpul de incarcare
- Early Stopping - oprirea antrenarii atunci cand performanta pe setul de validare nu mai se imbunatateste, prevenind overfitting-ul
- + Gradient Accumulation - acumularea gradientilor pe mai multe batch-uri mici inainte de actualizarea ponderilor, simuland un batch mai mare fara consum suplimentar de memorie.
- + Gradient Checkpointing - salvarea activarilor intermediare si recalcularea lor la backpropagation
- + EMA (Exponential Moving Average) - copie separata a modelului ale carei ponderi sunt actualizate treptat ca medie exponentiala a ponderilor curente din timpul antrenarii.
- + Dynamic quantization (cpu) - reducerea preciziei ponderilor in timpul inferentei pentru a scadea memoria folosita si a accelera executia

CONFIGURATIILE MODELELOR

Model	Img Size	Batch	Epochs	Optimizer	LR	Loss
U-Net	512	6	12	AdamW	1e-4	Dice + BCE
SegFormer-B0	512	4	12	AdamW	3e-5	Dice + BCE
DINOv2	518	2	6	AdamW	2e-5	Dice + BCE
SAM + head	1024	1	4	AdamW	1e-4	Dice + BCE
Mask R-CNN	512	1	6	AdamW	1e-4	Mask R-CNN losses
YOLO11n	640	8	20	AdamW	1e-3	Yolo losses

REZULTATELE MODELELOR

Model	Best val Dice	Best val IoU	Test Dice	Test IoU
SegFormer-B0	0.9256	0.9116	0.9466	0.9350
Mask R-CNN	0.9188	0.9042	0.9363	0.9201
U-Net	0.9159	0.9020	0.9306	0.9200
DINOv2	0.8706	0.8675	0.8953	0.8922
SAM + head	0.8418	0.8364	0.8755	0.8716

REZULTATELE MODELELOR

- YOLO11n:

Metric	Value
mAP50(B)	0.4825
mAP50-95(B)	0.2300
Latency	2.03ms
Throughput	491.50 img/s

Yolo a fost rulat ca baseline pt detectie, de aceea metricile sale nu sunt compatibile direct cu Dice/IoU. Totusi, punctul forte este viteza foarte mare.

SAM PROMPTING VS SAM HEAD

Model	Test Dice	Test IoU	Latency ms	Samples
SAM + Head	0.8755	0.8716	479.80	X
SAM Prompt Box	0.9027	0.8736	484.62	60
SAM Prompt Point	0.8577	0.8304	479.65	60

SAM se bazeaza pe prompting derivate din ground truth

Prompt Box = delimitarea obiectului printr-un dreptunghi care indica aproximativ zona ce trebuie segmentata.

Prompt Point = selectarea prin puncte a ce regiuni trebuie incluse sau excluse din segmentare.

TRADE-OFF ACCURACY VS COMPUTE

AMP = fp16 pentru fiecare model

Model	Test Dice	Min/Ep	Peak Memory mb	Latency ms	Throughput(img/s)	Params
SegFormer-B0	0.9466	10	291.23	12.70	78.75	3.72M
U-Net	0.9306	5	2535.6	19.98	50.05	24.44M
Mask R-CNN	0.9363	55	797.55	85.99	11.63	43.72M
DINOv2	0.8953	40	2351.892	130.81	7.64	88.13M
SAM + head	0.8755	60	8760.32	494.80	2.08	4.43M*

SegFormer-B0: cel mai bun trade-off accuracy si latentia

U-Net: cel mai bun baseline

SAM + head si DINOv2 sunt interesante metodologic, dar necesita prea multe resurse pentru utilizare practica in forma actuala

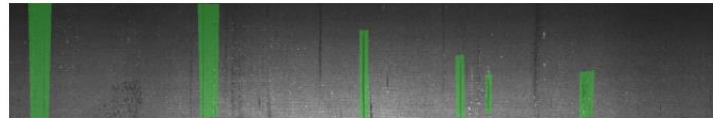
YOLO11n: date mentionate anterior, evident cel mai rapid

VIZUALIZAREA PREDICTIILOR

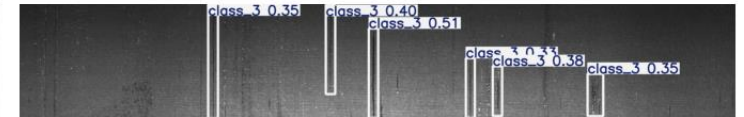
Image | severstal



Ground Truth

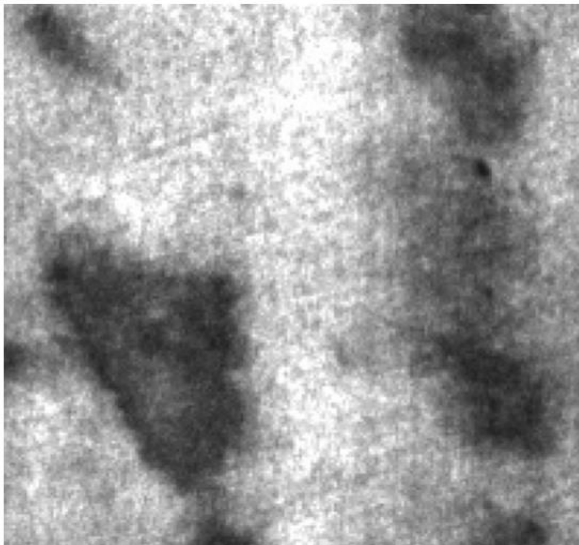


YOLO11 detect Prediction

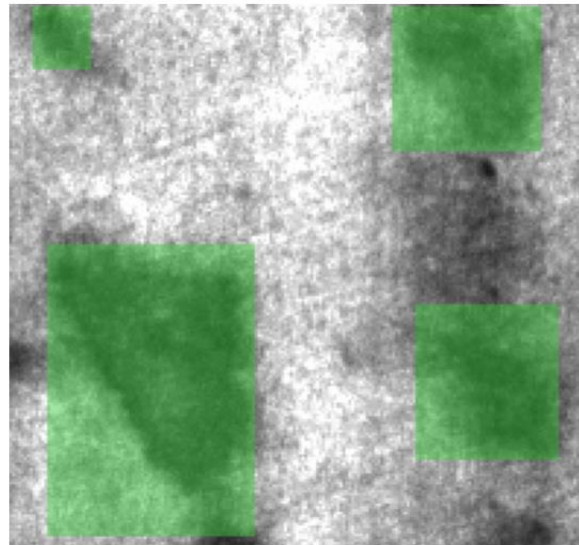


Mask R-CNN

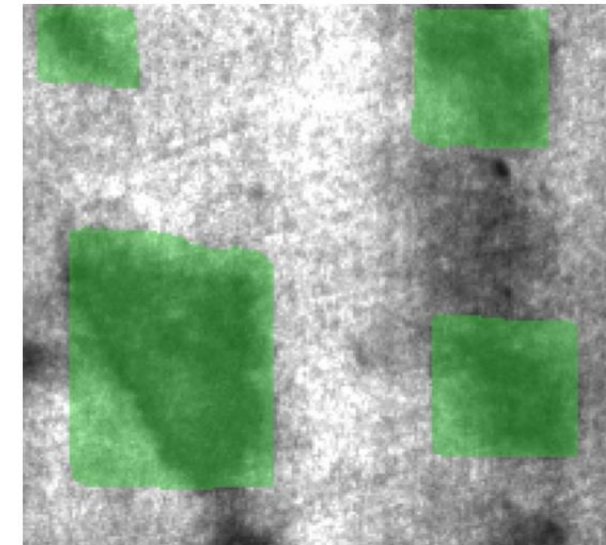
Image | neu



Ground Truth



Prediction



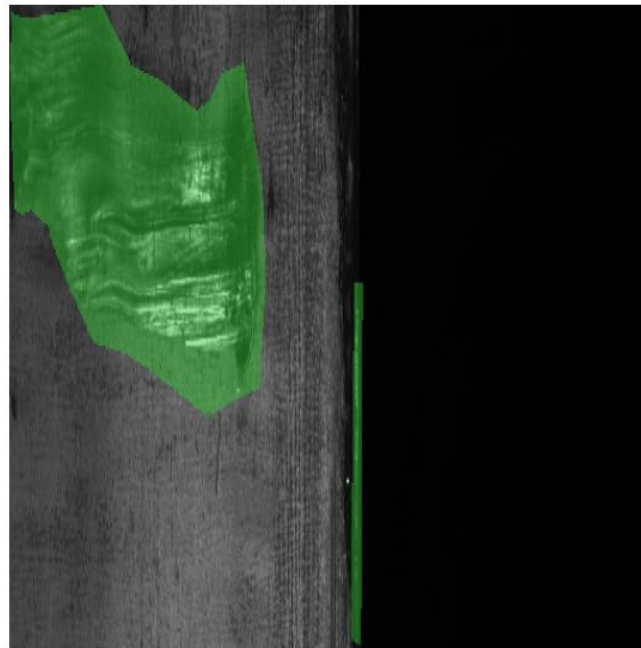
VIZUALIZAREA PREDICTIILOR

U-Net

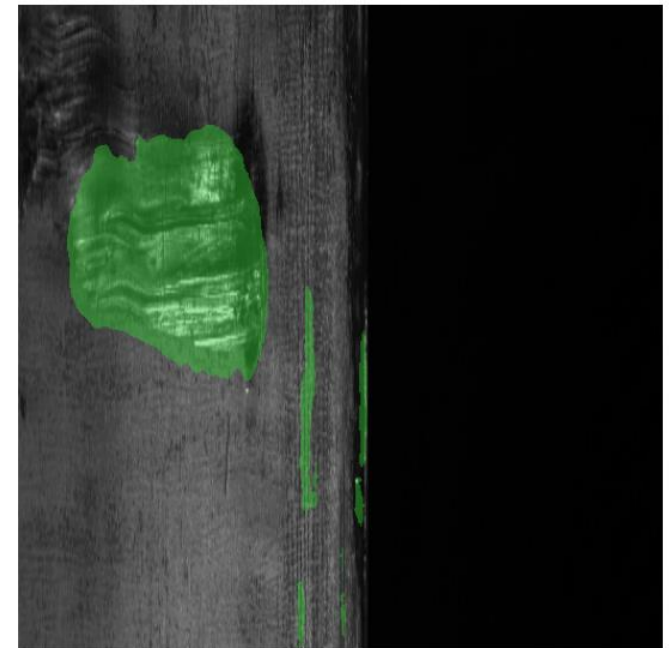
Image | severstal



Ground Truth



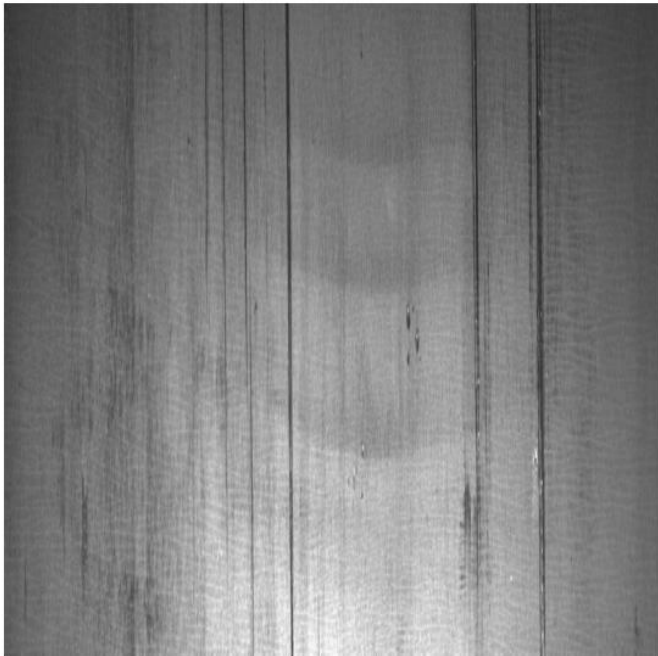
Prediction



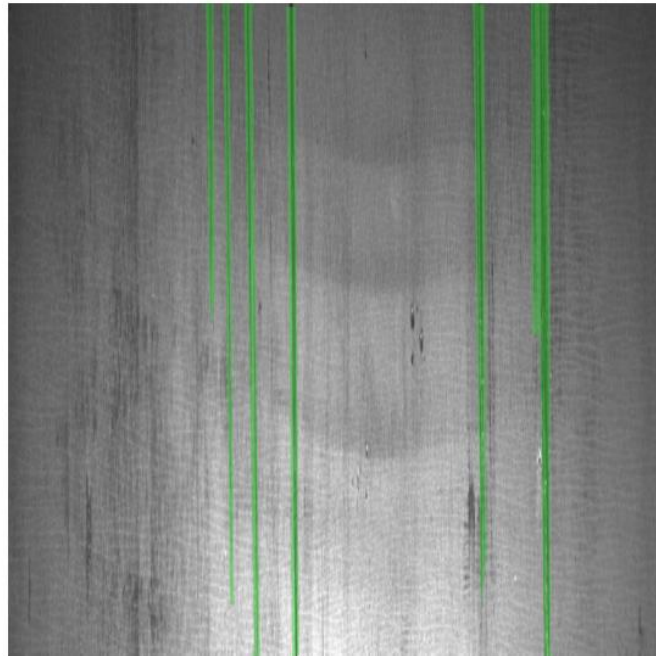
VIZUALIZAREA PREDICTIILOR

SAM

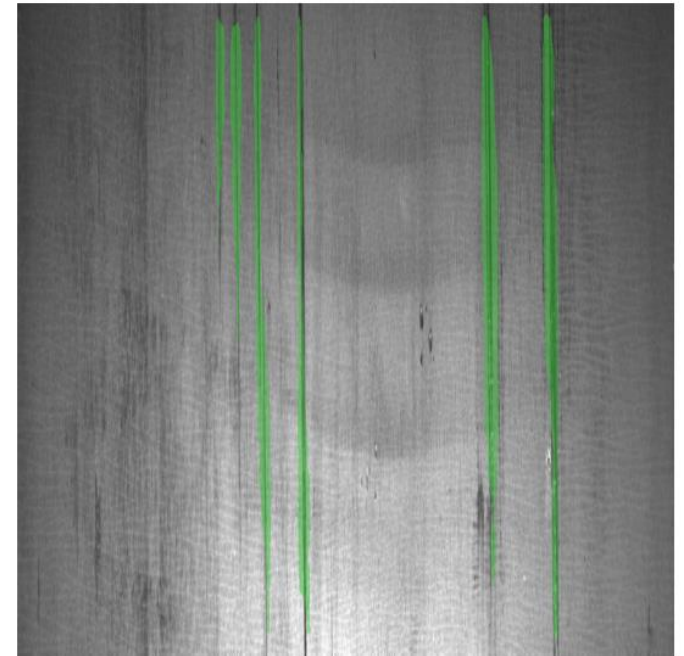
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Ground Truth



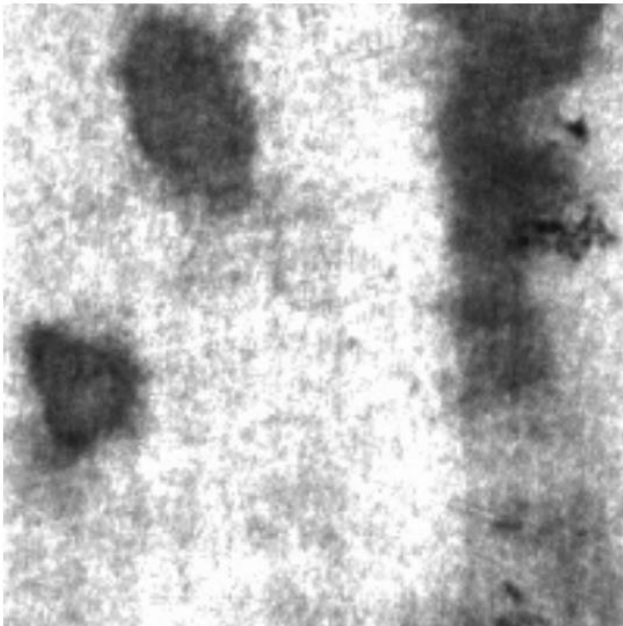
Prediction



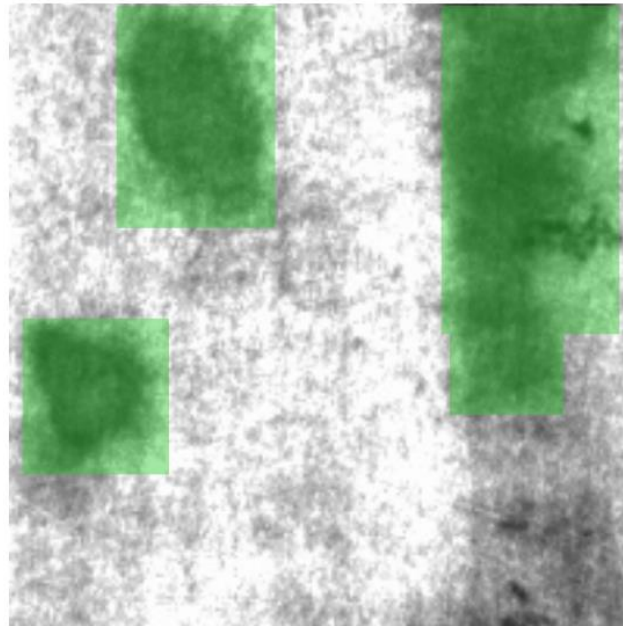
VIZUALIZAREA PREDICTIILOR

SegFormer-B0

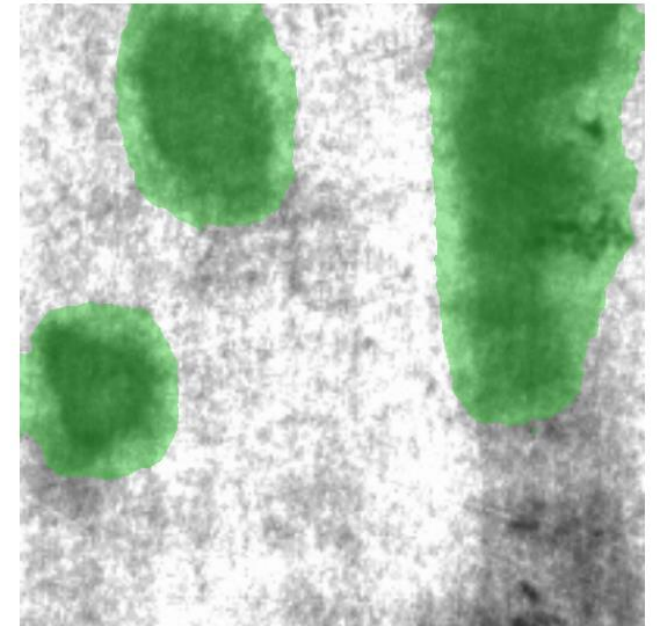
Image | neu



Ground Truth



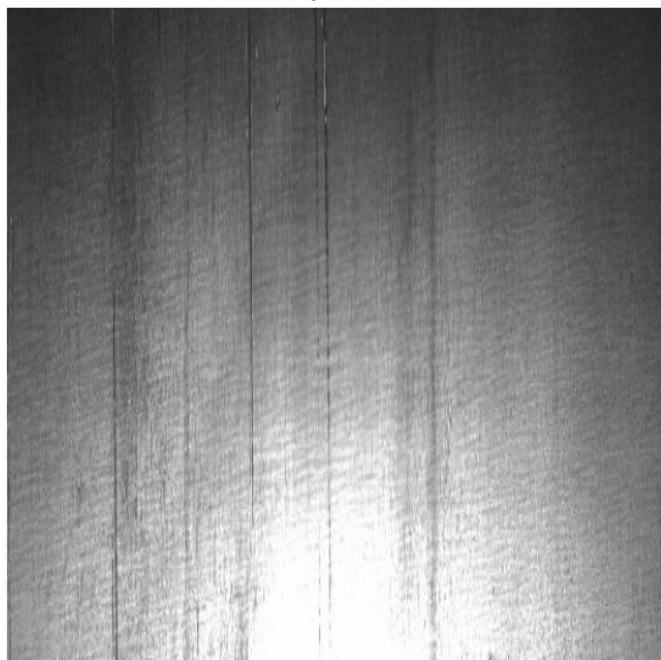
Prediction



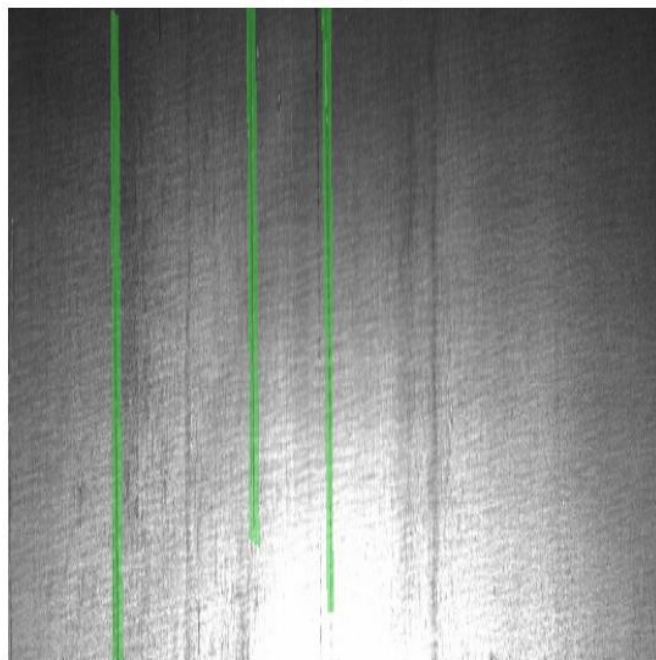
VIZUALIZAREA PREDICTIILOR

DINOv2

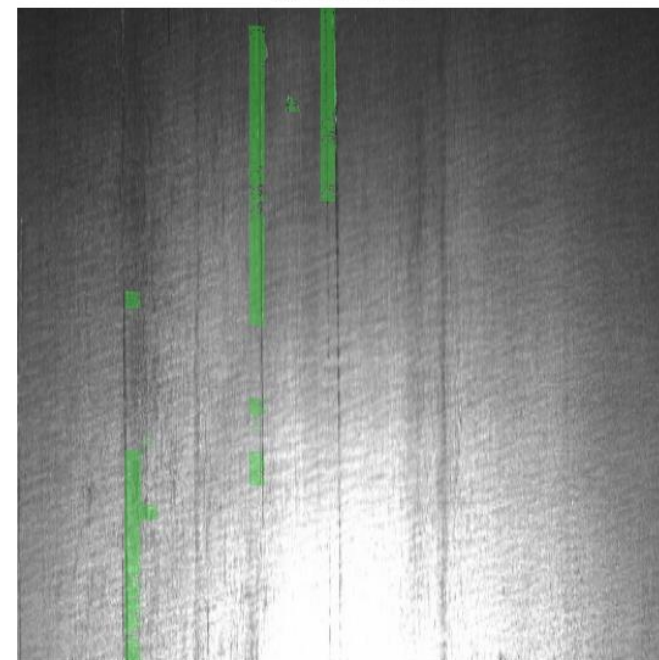
Image | severstal



Ground Truth



Prediction



CONCLUZII

- Vizual si numeric, modelele surprind bine zonele delimitate de defect.
- Totusi, in contextul nostru, DINOv2 si SAM + head nu au justificat costul computational.
- In schimb, SAM prompting prin box a fost superior SAM + head, care a necesitat finetunare
- Mask R-CNN ofera rezultate calitative, dar la un cost mai mare. E recomandat sa nu fie folosit in real-time.
- U-Net ofera un baseline solid, iar daca este adaptat la o versiune lightweight, se poate incerca utilizarea sa in real-time.
- YOLO, in setup-ul curent, ar trebui folosit doar cand viteza este requirement-ul principal.
- In final, am observat ca SegFormer-B0 a fost modelul cu cel mai bun compromis intre performanta si cost.

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- YOLO: <https://docs.ultralytics.com/>
- Vit: https://github.com/google-research/vision_transformer
- SegFormer-B0: <https://arxiv.org/abs/2105.15203>
- Sam: <https://github.com/facebookresearch/segment-anything>
- Dino: <https://github.com/facebookresearch/dinov2>

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